Generating a 1000-Member Ensemble of Integrated Water Vapor Transport Forecasts with Diffusion

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ABSTRACT: Ensemble weather predictions can be critical toward making informed decisions for operations that require risk assessment. The traditional approach toward creating an ensemble often involves making slight perturbations to initial conditions and model numerical solvers and physics to account for forecast uncertainty. This study instead uses diffusion, a form of generative artificial intelligence (AI), to make ensemble forecasts consisting of 1000 members from single-member deterministic dynamical Western Weather Research and Forecasting (West-WRF) Model forecasts over the North Pacific. After it has been trained, the diffusion ensemble can make predictions at high speeds and at a low computational expense while improving the skill of medium-range integrated water vapor transport (IVT). It has high performance when evaluated by binned ranked histograms and outperforms operational ensemble forecasts [Global Ensemble Forecast System (GEFS) and European Centre for Medium-Range Weather Forecasts (ECMWF)] in continuous ranked probability score, binned spread to root-mean-square error ratio, reliability diagrams, and Brier skill scores. The diffusion ensemble further demonstrated utility in representing real outcomes of extreme destructive events with realistic high-quality images.

SIGNIFICANCE STATEMENT: This work explores a novel method of creating probabilistic weather forecasts. By harnessing diffusion, a form of generative artificial intelligence, a large ensemble of medium-range forecasts of integrated water vapor transport was created using a deterministic forecast as input. The diffusion ensemble had better performance than several operational weather models based on a range of probabilistic skill metrics.

KEYWORDS: Atmospheric river; Ensembles; Numerical weather prediction/forecasting; Artificial intelligence; Machine learning

1. Introduction

Flooding, drought, and available water resources on the west coast of the United States are highly sensitive to strong fluctuations in precipitation activity, which can often be driven by powerful vapor transport events known as atmospheric rivers (ARs). A single AR event that occurred in Northern California in 2017 caused 188 000 people to evacuate and over $1 billion in damages (Henn et al. 2020). As the planet warms due to anthropogenic forcing, extreme ARs are projected to become more severe (Corringham et al. 2022; Higgins et al. 2025), likely leading to more lives at risk and damages to infrastructure. In some cases, such as in late 2022, AR activity can also lead to the end of long-lasting drought (Purohit 2025). To navigate the challenges posed by a volatile climate in an area with large populations and rich in agriculture, it is critical to predict incoming vapor transport accurately.

Skillful weather forecasts are an essential technology for managing everyday decisions and preparing for potentially dangerous events. While forecasts continue to become more reliable as observations, models, and assimilation systems improve, they are limited by chaos (Lorenz 1963; Durran and Gingrich 2014; Palmer et al. 2014), which causes even small errors to inevitably grow exponentially with time until forecasts eventually become useless. The chaotic nature of the atmosphere results in uncertain weather forecasts, which need to be reliably quantified to increase the value of the prediction. Forecast models often account for chaos and uncertainty by making ensemble predictions. The model is run multiple times to represent multiple possible outcomes to make an ensemble prediction. Reliable ensembles span possible outcomes that are equally as likely to occur a priori as the true outcome. Traditionally, ensemble members have perturbed initial conditions, parameterization schemes, and/or varied model physics. Although these methods are effective, they can be computationally expensive and, in many cases, struggle to make the dispersion of the ensembles properly reflect the uncertainty of the forecast.

Alternative methods can be used to generate ensembles from one prediction based on deterministic physics and a historical dataset of predictions from that system and observations of the predictand [e.g., as the analog ensemble presented by

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TABLE 1. All five datasets used in the study.

Dataset	Purpose	Spatial resolution	Ensemble members
ERA5	Training and evaluation	0.25° × 0.25°	—
West-WRF	Training, inference, and benchmark	0.25° × 0.25° (regridded from 9 km × 9 km during preprocessing)	1
ECMWF EPS	Benchmark	0.5° × 0.5°	50
NCEP GEFS	Benchmark	0.5° × 0.5°	20
Diffusion	Output result	0.25° × 0.25°	1000

Delle Monache et al. (2013)]. Although analog ensembles are highly computationally efficient and generally reliable, their ensemble size and performance are confined by a limited historical record. Another effective technique is to blend multiple forecast models to create ensembles (Johnson and Swinbank 2009; Woodcock and Engel 2005), which requires multiple processing steps that carefully adapt the scheme to the desired product. In addition, quantile regression methods (Bjørnar Bremnes 2004) have been applied to many weather variables and can produce reliable marginal distributions with some limitations, including spatial coherence and temporal consistency (Taillardat et al. 2016; Rasp and Lerch 2018). Although each existing method is valuable to the forecasting community, it is possible to continue to advance ensemble predictions by exploring other possible approaches.

Generative artificial intelligence (AI) is rapidly emerging as an effective tool for a variety of applications in weather prediction. Generative adversarial networks (GANs) train two neural networks against each other, with one that creates new data and another that predicts whether the new data are real or fake. They have been used for postprocessing, downscaling, and ensemble generation (Bihlo 2021; Harris et al. 2022; Sha et al. 2024). Variational autoencoders (VAEs) are an alternative image generation method that samples latent vectors from a distribution to create synthetic data. They have demonstrated utility in making forecasts, data assimilation, and simulating extreme events (Kwok and Qi 2021; Melinc and Zaplotnik 2024; Szwarcman et al. 2024). While both types of deep generative models are useful for completing various tasks that can enhance weather prediction, they still leave room for improvement if the goal is the generation of well-calibrated ensembles (Sha et al. 2024; Chen et al. 2025).

Diffusion was first introduced in Sohl-Dickstein et al. (2015) and used concepts from nonequilibrium thermodynamics to synthesize high-quality images from white Gaussian noise fields. Useful applications of diffusion include the emulation of fluid dynamical systems (Luo et al. 2024; Molinaro et al. 2025; Pedersen et al. 2025), with some studies specifically focusing on generating large-ensemble climate simulations (Meuer et al. 2024), creating autoregressive ensemble forecasts (Price et al. 2023), and postprocessing ensemble forecasts (Li et al. 2024). Ho et al. (2020) introduced denoising diffusion probabilistic models (DDPMs) as a method of generating high-quality images that laid the foundation for many subsequent advances of diffusion-based generative models. It starts with a forward process in which Gaussian noise is gradually added to a clean image in a series of steps until the new image is corrupted entirely. The model is trained to learn the amount of noise that was added at each step, minimizing a mean-square error (MSE) loss function between the predicted noise and the actual noise. Once the model is trained, the reverse process predicts the noise in the image in a series of multiple steps and subtracts it, eventually resulting in a new clean image.

In this study, we implement the denoising diffusion implicit model (DDIM) to train a diffusion model that can generate a 1000-member ensemble of integrated water vapor transport (IVT) forecasts over the North Pacific. We use a conditional diffusion model, with the input condition being deterministic medium-range forecasts. Conditioning on the deterministic forecast restricts the model to generate atmospheric states consistent with the current forecast situation, rather than sampling from the full range of states found in the climatology. The ensemble members are not connected temporally and do not attempt to capture the evolution of flow. The model is trained using data from both deterministic forecasts and a reanalysis product, which allows the ensemble to reduce forecast errors. Once the model is trained, the ensemble is generated from deterministic forecasts at a relatively low computational cost. This approach may be particularly useful for forecasts of extreme events with high uncertainty. For both physics-based and data-driven-based methods of ensemble generation, a large ensemble may be beneficial to better represent the right tail of the IVT distribution, resulting in both a more accurate prediction and better uncertainty quantification of extreme events (Tempest et al. 2023; Mahesh et al. 2024; Delle Monache et al. 2025).

The remainder of this paper is structured as follows: Section 2 describes the diffusion model and the data used to train it. Section 3 describes the methods used to implement the diffusion model. Section 4 discusses the performance and reliability of the ensemble and provides examples of cases in which it captures uncertainty. Section 5 gives concluding remarks and a discussion of possible future directions of the work.

2. Data

All five datasets used in the study are summarized in Table 1. We train the diffusion model to predict noise with water years 1986–2015 for the months of December–March from the fifth generation European Centre for Medium-Range Weather Forecasts atmospheric reanalysis (ERA5). A version of the National Center for Atmospheric Research Western Weather

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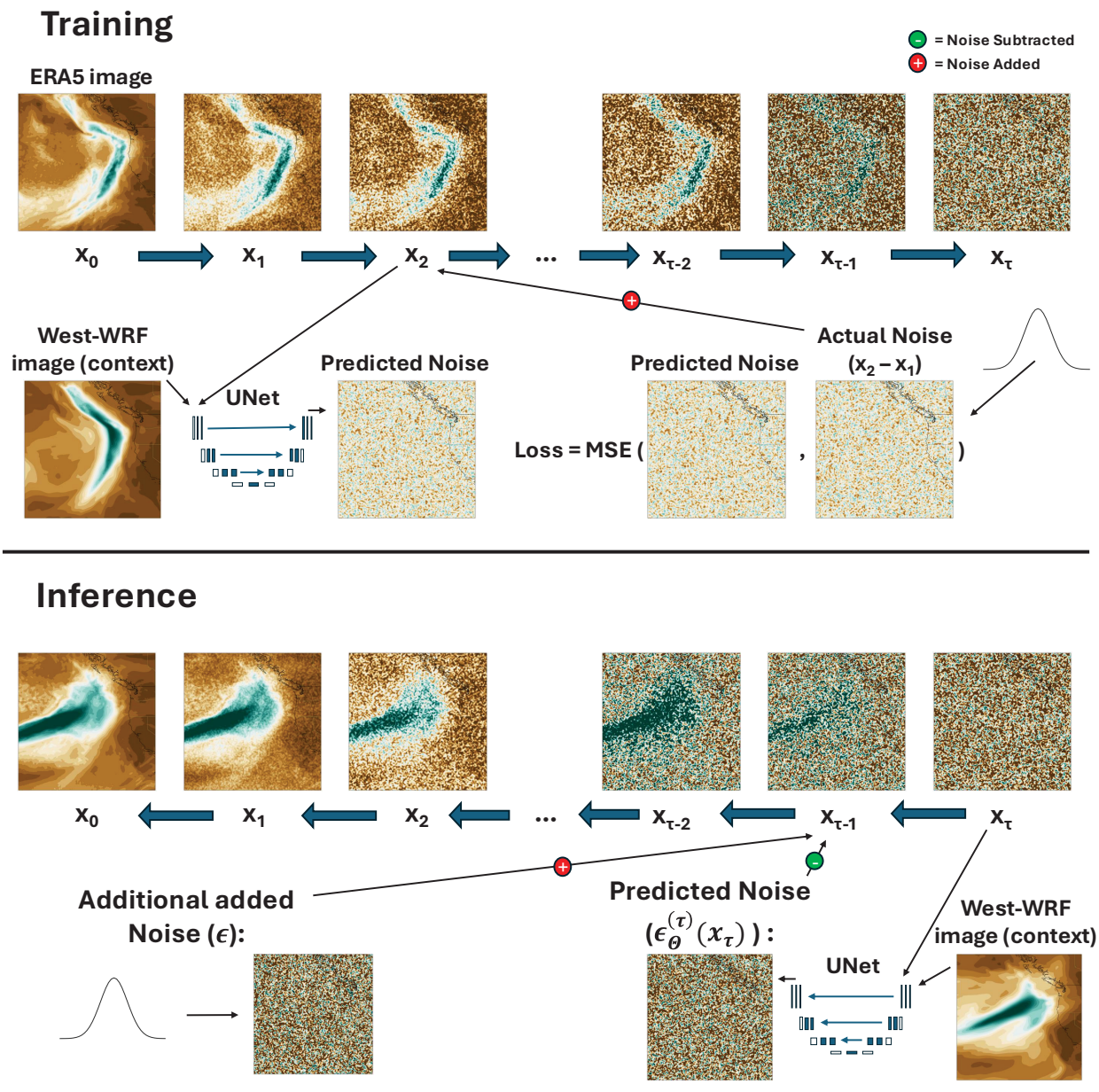

FIG. 1. A visual representation of the training and inference processes. The training process gradually adds noise to the image and predicts the noise added. The inference process uses a trained CNN to predict the noise in the image and subtract it while adding some additional noise to favor sample diversity. The deterministic condition is concatenated into the CNN to serve as context. The model is trained separately for each lead time.

Research and Forecasting (West-WRF) Model functions as the input condition for noise predictions in both the training and inference processes. Water years 2016–18 were used as test data, and ERA5 was used as "ground truth" for verification. Water year 2019 was used for the validation set. Four NVIDIA A100 graphics processing units (GPUs) were used for training and one NVIDIA A100 GPU was used for inference. Once training was completed, each 1000-member ensemble forecast took less than 2 h to generate for a single lead time. The diffusion model was trained for forecasts initialized at 0000 UTC with lead times of 3, 4, and 5 days ±3 h. While it is possible to train the model for additional lead times, constraints on time and computational resources restricted this study to days 3–5. Each lead time was trained for roughly 10 000 epochs and used roughly 21 600 images. ERA5 uses a 0.25° × 0.25° horizontal grid resolution, and a conservative regridding scheme was used to make West-WRF data have a matching resolution. The spatial domain spanned much of the northeast Pacific and included the west coast of North America, going from 28.25° to 60°N and from 115° to 146.75°W.

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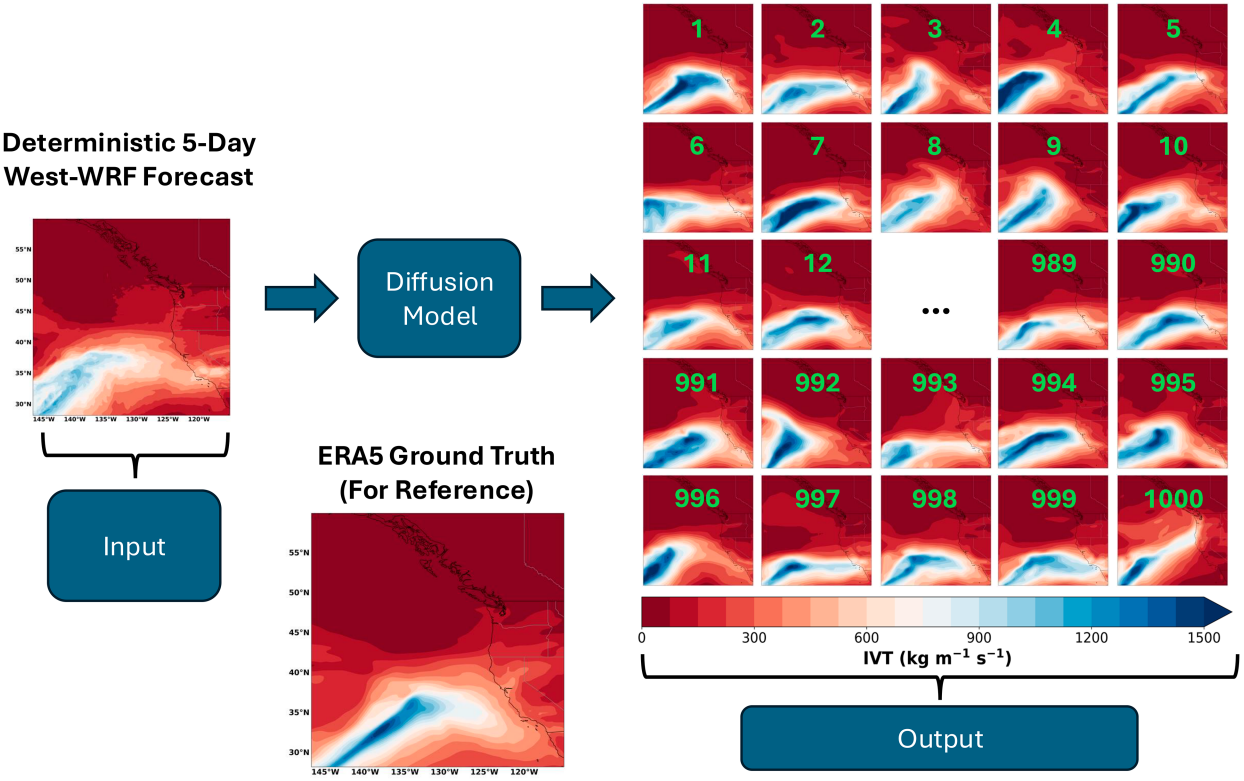

FIG. 2. An example of an AR taking place at 0000 UTC 7 Feb 2017 that was involved in causing extreme flooding in Northern California. The diffusion model creates an ensemble from a 5-day deterministic forecast as conditional input. Twenty-four random members out of 1000 are shown here. ERA5 IVT is also shown at the same valid time as the forecast for reference.

The performance of the diffusion ensemble is compared to that of the European Centre for Medium-Range Weather Forecasts (ECMWF)'s Ensemble Prediction System (EPS) and the U.S. National Centers for Environmental Prediction (NCEP)'s Global Ensemble Forecast System (GEFS). EPS Cycle 49r1 (implemented in 2024) and GEFS version 12 (implemented in 2020) were used for comparisons. EPS and GEFS had native resolutions of 9 and 25 km, respectively. The archived reforecasts used in this study for both models were stored with a spatial resolution of 0.5° × 0.5°. EPS used 50 ensemble members, and GEFS used 20 ensemble members. Data from both models in the study only consisted of forecasts from the same dates and spatial domain as the test set.

3. Methods

Diffusion models generate samples by learning to reverse a gradual noising process. We provide a brief overview here and refer the reader to Ho et al. (2020) and Song et al. (2020) for further details. Intuitively, the forward process starts from a real atmospheric field and adds a small amount of Gaussian noise at each of many steps, until the field is indistinguishable from pure noise. A convolutional neural network (CNN) is trained to estimate the noise added at each step. To generate a forecast, the reverse process runs in the opposite direction. Starting from pure noise, the network iteratively predicts and removes noise to recover a realistic field. Because we condition the network on the deterministic West-WRF forecast at every step, the recovered fields are constrained to states consistent with that forecast rather than the full climatology, and repeating the reverse process from different noise samples yields an ensemble. This procedure is illustrated in Fig. 1.

The forward process produces a noised field x_τ at time step τ from the original field x_0 as

$$x_\tau = \sqrt{\alpha_\tau} x_0 + \sqrt{1 - \alpha_\tau}\varepsilon, \text{ where } \varepsilon \sim \mathcal{N}(0, I), \qquad (1)$$

where α_τ is a variance parameter between 0 and 1 that decreases as τ increases and ε is the standard Gaussian noise. We use 1500 steps with a linear scheduler, so that α decreases by a constant amount at each step. This choice promotes greater sample diversity and, in turn, larger ensemble spread. The noise is predicted by a convolutional U-Net (Ronneberger et al. 2015). To make the generated states depend on the deterministic forecast, we concatenate the West-WRF condition into the U-Net input, so that contextual information is incorporated across all layers of the encoder and decoder.

For sampling, we use the DDIM formulation (Song et al. 2020), which improves efficiency by using a non-Markovian

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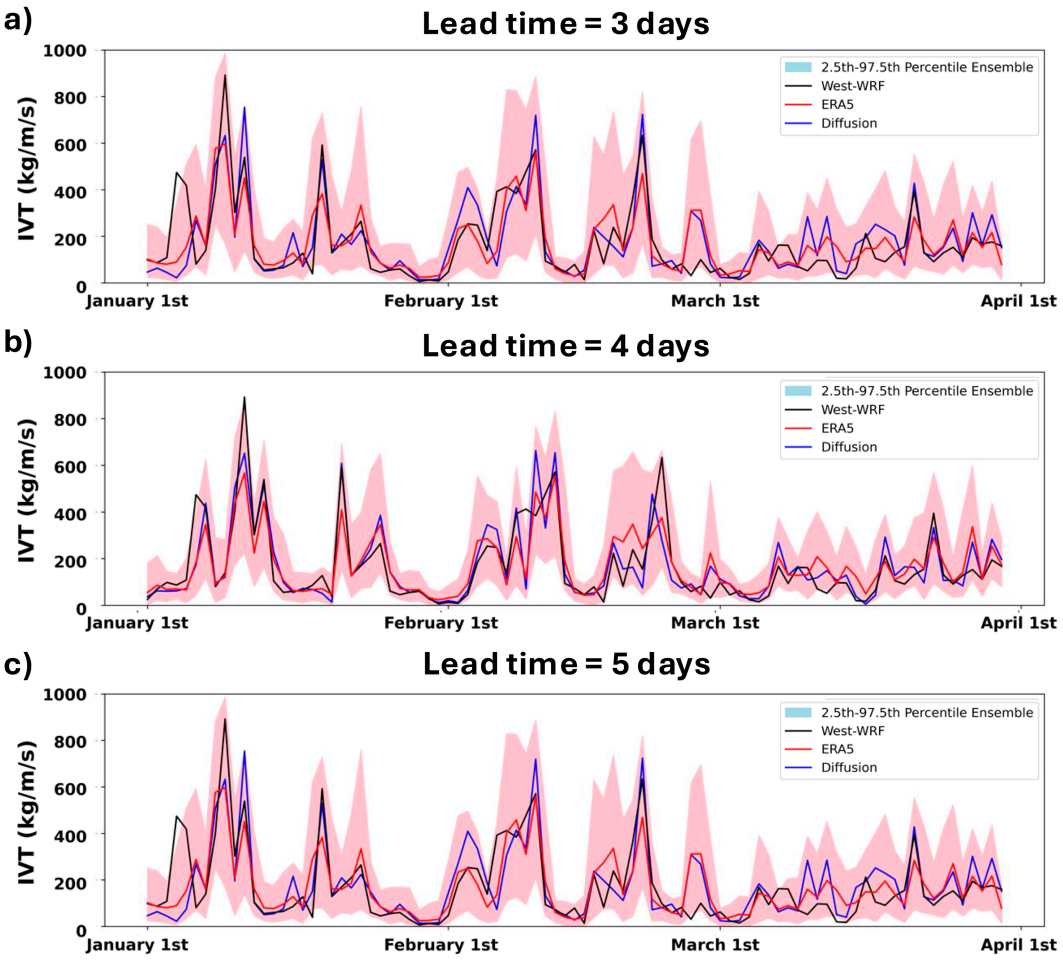

FIG. 3. (a)–(c) Time series of IVT (37.5°N, 122.5°W, San Francisco) for lead times of 3–5 days from 1 Jan 2017 to 31 Mar 2017. Values of IVT are shown from West-WRF (blue), ERA5 (black), the diffusion ensemble mean (red), and the range of the 2.5th percentile to 97.5th percentile diffusion ensembles (light red).

reverse chain. At each reverse step, a small amount of noise is reintroduced to increase sample diversity and ensemble spread. The standard deviation of this injected noise is

$$\sigma_\tau = \eta \sqrt{\left(1 - \frac{\alpha_\tau}{\alpha_{\tau-1}}\right)\left(\frac{1-\alpha_{\tau-1}}{1-\alpha_\tau}\right)}\,, \tag{2}$$

where η controls the magnitude of the added noise in the denoising process; we set $\eta = 1$. A single reverse step from x_τ to $x_{\tau-1}$ is then

$$x_{\tau-1} = \sqrt{\alpha_{\tau-1}}\frac{x_\tau - \sqrt{1-\alpha_\tau}\varepsilon_\theta^{(\tau)}(x_\tau)}{\sqrt{\alpha_\tau}} + \sqrt{1-\alpha_{\tau-1}-\sigma_\tau^2}\varepsilon_\theta^{(\tau)}(x_\tau) + \sigma_\tau\varepsilon_\tau, \tag{3}$$

where $\varepsilon_\theta^{(\tau)}(x_\tau)$ is the noise predicted by the U-Net at step τ. The first term is an estimate of the original field x_0, obtained by rearranging Eq. (1) with the predicted noise in place of ε and scaling by $\sqrt{\alpha_{\tau-1}}$. The second term redirects a controlled fraction of the predicted-noise direction back toward x_τ; the third term, $\sigma_\tau\varepsilon_\tau$ with $\varepsilon_\tau \sim \mathcal{N}(0, I)$, is the freshly sampled noise that drives ensemble diversity.

4. Results

a. *Example case*

Once the diffusion model is trained, it can be used to create an ensemble of any size when given a deterministic forecast as an input condition. Here, we create a 1000-member ensemble to demonstrate the ability of the model to predict an extreme event (Fig. 2). This example took place on 7 February 2017, which marked the beginning of the Oroville Dam crisis (Henn et al. 2020; Vahedifard et al. 2017). The West-WRF 5-day forecast predicted a large plume of IVT upstream of the coast that spanned across much of the northeast Pacific. In contrast, ERA5 showed an AR that, while in a similar location, was stronger and had a more filamentary shape. The diffusion

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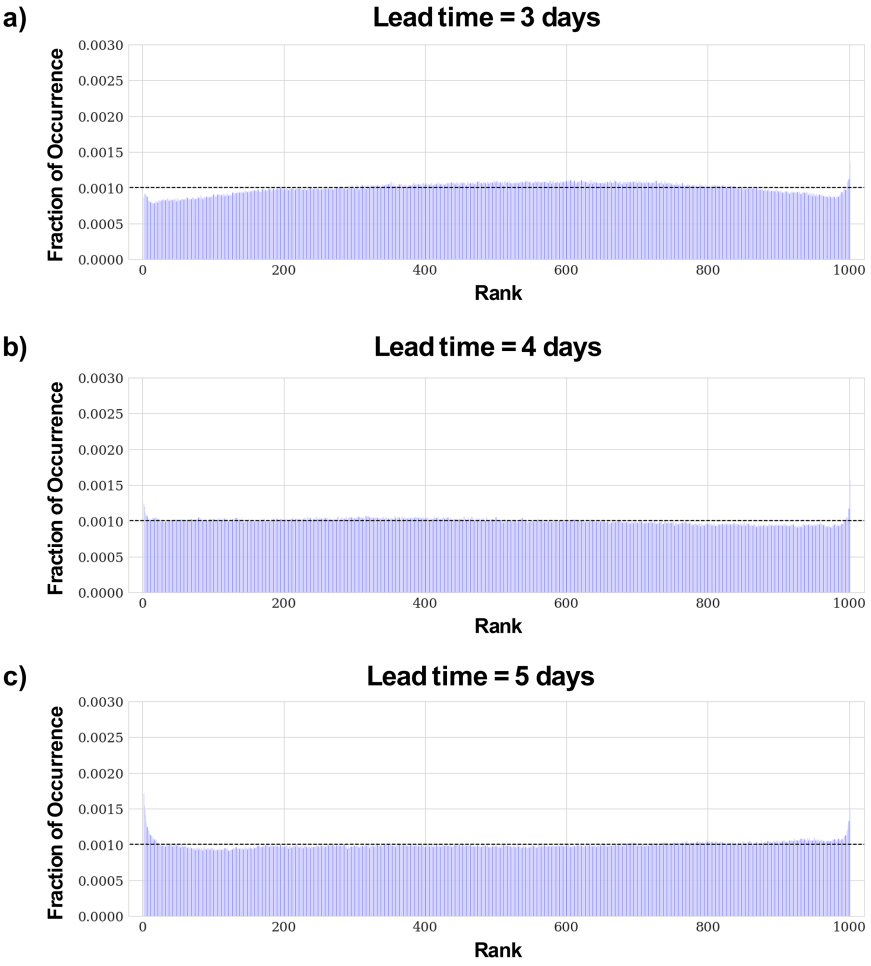

FIG. 4. (a)–(c) Binned ranked histograms for the diffusion ensemble across the entire spatial domain for lead times spanning from 3 to 5 days in the test set. The *x* axis represents the rank of ERA5 among the diffusion ensemble, and the *y* axis represents the frequency of each rank. The horizontal dotted line marks the frequency at which each rank would occur in a perfectly calibrated probabilistic forecast.

model generated a variety of different IVT fields, many of which captured the general structure and magnitude of the AR that West-WRF did not. An advantage of using diffusion to generate possible states of ARs is that it avoids the double-penalty problem. When loss functions are used to minimize the difference between values in the forecast and ground truth, the resulting model can create blurry fields that decrease MSE but are not realistic. By using the loss function to predict noise, the diffusion model generates highly realistic fields of ARs. Forecast activity, which measures the smoothness of numerical weather forecasts (Ben-Bouallegue et al. 2023), showed that the diffusion ensemble had even sharper images than West-WRF did (Fig. S1 in the online supplemental material). When the diffusion ensemble mean was compared to ERA5, the root MSE (RMSE) was 16.5% lower than that of West-WRF. In this case, the diffusion ensemble could have provided some useful insight into the prediction of characteristics associated with the AR.

b. Bias statistics

Across all grid points, the mean magnitude of bias for West-WRF was 7%–8% in both the training set and the test set. In contrast, when the diffusion postprocessing ensemble was utilized, the mean magnitude of the bias was 3% lower than that of West-WRF in the test set. We further selected the point (37.5°, 122.5°), which is roughly the location of San Francisco. This point was chosen because it is a densely populated area with a large economy that experiences frequent AR activity. Some biases existed at this point, with West-WRF being 5% higher than ERA5 in the test set and 2% higher in the training set. The diffusion ensemble-mean IVT was 1% higher than ERA5 IVT in the test set.

During the winters (January–March) of 2016–2018, the ERA5 ground-truth IVT fell outside the 2.5th–97.5th percentile range of the diffusion ensemble in 7%, 6%, and 7% of cases, respectively, where each case is a single forecast valid time. Of the cases in which the ERA5 value fell outside this

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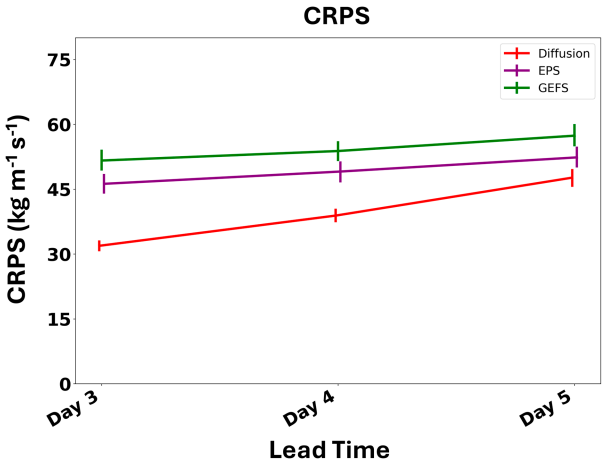

FIG. 5. CRPS of GEFS, EPS, and the diffusion ensemble at lead times of 3, 4, and 5 days.

range, 58% exceeded the 97.5th percentile and 42% fell below the 2.5th percentile, indicating a slight dry bias in the tails of the ensemble distribution.

c. *Extreme season*

Based on a 70-yr record from 1948 to 2017, the winter of 2017 [January–March (JFM)] had the most AR-related IVT landfalling over California, exceeding the mean by three standard deviations (Gershunov et al. 2017). This barrage of storms effectively ended a multiyear drought. Because this season lies within the test set and represents the most extreme AR activity in the record, we use it to examine ensemble behavior under highly anomalous conditions (Fig. 3).

Throughout this season, the ERA5 IVT, including its largest peaks, remained within the 2.5th–97.5th percentile range of the diffusion ensemble in 94% of cases, indicating that the ensemble reliably captured the observed uncertainty even during an extreme winter. As an ensemble mean, the diffusion mean does not itself reproduce the sharpest peaks, but the spread of members spans them. The mean IVT at the selected point during this season was 168.3 kg m^{-1} s^{-1} in ERA5, 176.4 kg m^{-1} s^{-1} in West-WRF, and 171.2 kg m^{-1} s^{-1} in the diffusion ensemble. Taking the training-period mean at this point (114.8 kg m^{-1} s^{-1} in ERA5, 109.2 kg m^{-1} s^{-1} in West-WRF) as a climatological reference, the diffusion ensemble mean during this extreme season fell between the West-WRF input forecast and climatology, which is consistent with a model that moderates the input condition toward typical conditions. The RMSE of the diffusion ensemble mean was also 8% lower than that of West-WRF.

d. *Probabilistic forecast skill*

Binned ranked histograms, continuous ranked probability score (CRPS), and binned spread versus RMSE (Talagrand 1999; Haiden et al. 2024) were used to understand the quality of the diffusion ensemble calibration (Figs. 4–6). CRPS, RMSE, and ensemble spread were first calculated at each grid point for each forecast. Mean CRPS was calculated from the mean over the entire spatial domain (Fig. 5). Binned spread versus RMSE was calculated by separating the ensemble spread values from all forecasts over the entire spatial domain into 10 bins of equal size and calculating the mean RMSE for each bin (van den Dool 1989; Wang and Bishop 2003). All evaluation metrics include data from the entire spatial domain and for the entire test set (water years 2016–18). In the binned ranked histograms (Fig. 4), the value of ERA5 IVT is ranked against each of the 1000 diffusion ensemble members at each grid point and for each forecast. This results in a distribution of frequencies of all ranks from 1 to 1000. In a perfectly calibrated ensemble, each rank would occur an equal number of times, resulting in a flat-ranked histogram (Anderson 1996; Hamill and Colucci 1997; Talagrand 1999). The large number of ensemble members improves the resolution of the ranked histogram at the tails (Fig. S2). There was high variability among ranked histograms across training epochs (not shown). The selected models were chosen based on the quality of calibration in the validation set. Each selected model version was then run on the test set for evaluation. The resulting binned

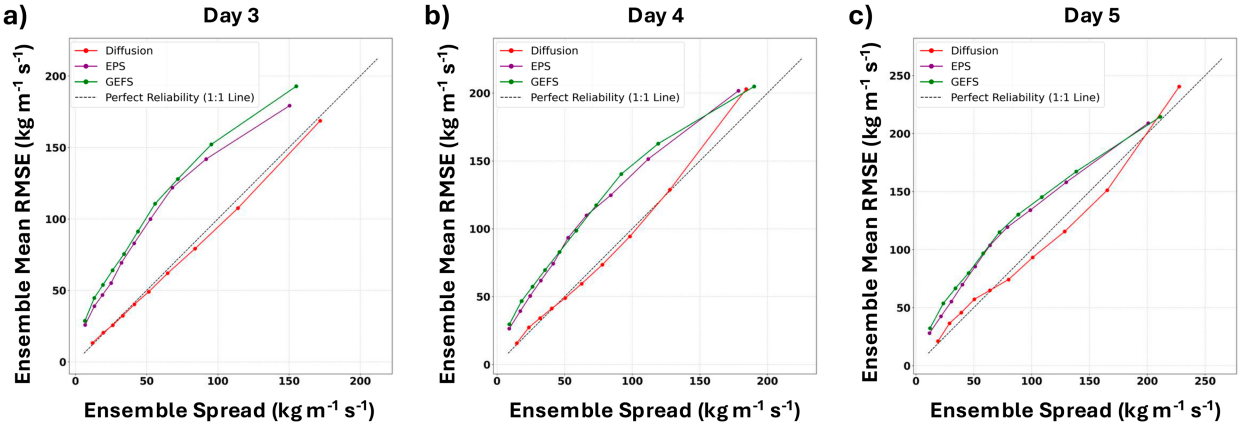

FIG. 6. (a)–(c) Binned-spread skill plots of GEFS, EPS, and the diffusion ensemble for lead times of 3–5 days. The dashed line represents the relationship between RMSE and binned spread for a perfect ensemble forecasting system.

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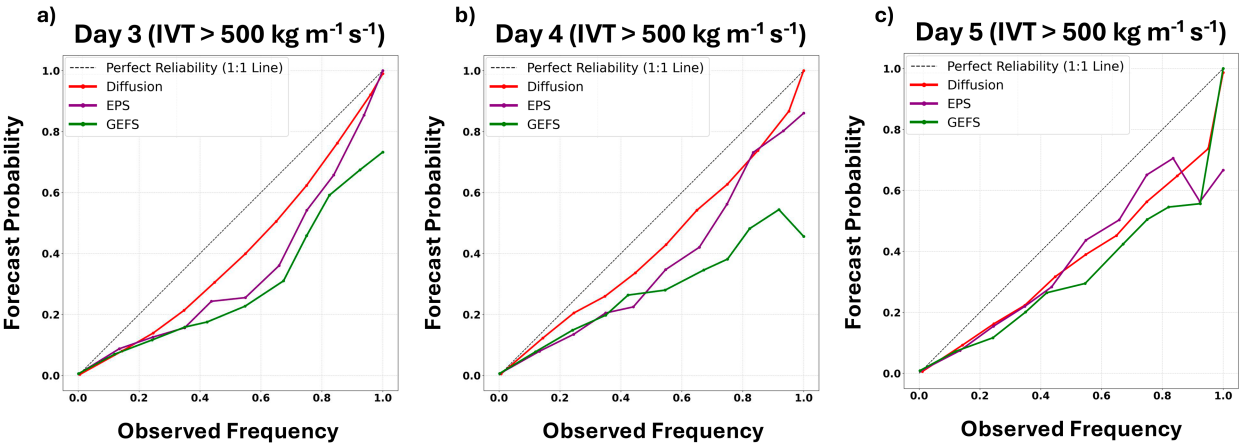

FIG. 7. (a)–(c) Reliability diagrams of predictions of IVT exceeding 500 kg m^{-1} s^{-1} in GEFS, EPS, and the diffusion ensemble at lead times of 3–5 days. The dotted line represents the relationship between forecast probability and observed frequency in a perfectly calibrated system.

ranked histograms were mostly flat, indicating that it is a well-calibrated system. CRPS and binned spread versus RMSE are also used to show the ensemble calibration and sharpness of both GEFS and EPS (Figs. 5 and 6). The diffusion ensemble outperforms both models in both metrics at all three lead times. The diffusion ensemble is generally well calibrated with some imperfections, which were primarily caused by the ensemble being slightly underdispersive at 4 and 5 days while being slightly overdispersive at 3 days. It is possible to further improve the calibration by using untested epochs or by further optimizing the number of steps or the amount of noise added during inference η for each individual lead time.

It is critical to understand the ability of the diffusion ensemble to predict conditions associated with extreme events. To do this, we first show reliability diagrams for all forecasts in the test set with an IVT threshold of 500 kg m^{-1} s^{-1} (98th percentile) (Fig. 7). At day 3 and day 4, the reliability diagrams show that the observed frequencies of IVT exceeding 500 kg m^{-1} s^{-1} are more closely aligned with the forecast probabilities in the diffusion ensemble than they are in EPS and GEFS. Day 5 shows more of a mixed result, with EPS having the best performance at several frequencies and the diffusion ensemble having better performance at others.

Figure 8 shows Brier skill scores (Figs. 8a–c) and their components, reliability (Figs. 8d–f), and resolution (Figs. 8g–i) in all three models for thresholds ranging from 250 to 500 kg m^{-1} s^{-1} in the test set. The diffusion ensemble has the highest Brier skill scores and resolution for all three lead times and at all thresholds. It also outperforms the operational models in reliability for days 3 and 4, with a mixed result in day 5 (Fig. 7). This indicates that at day 5, the diffusion model has higher performance in distinguishing between cases in which extremes are likely or unlikely but has slightly diminished ability to match probabilities to observed outcomes. This is consistent with Fig. 4, which shows that the diffusion ensemble is slightly underdispersive at day 5, and Figs. 5 and 6, which show that the performance of day 5 is slightly weaker than that of days 3 and 4. Despite this limitation, the diffusion ensemble performs well overall in predicting IVT extremes relative to EPS and GEFS.

e. *Deterministic forecast skill*

Deterministic forecast skill was also evaluated for each lead time (Fig. 9). The anomaly correlation coefficient (ACC) and RMSE were calculated by comparing West-WRF to ERA5 and the diffusion ensemble mean to ERA5. Overall, diffusion improves the RMSE by 3.9% and ACC by 2.6% over West-WRF. Bootstrapping was used to evaluate the significance of the difference in forecast skill. The differences were significant at the 95% level for ACC at all three lead times and for RMSE at day 4. The differences in forecast skill were not uniform in space. Some areas in the western part of the domain showed West-WRF having higher forecast skill than the diffusion mean. However, diffusion improved forecast skill in most areas and improved skill the most on and near the U.S. West Coast. Differences across space for each individual lead time are shown in Fig. S3. The diffusion ensemble mean had higher forecast skill than the EPS ensemble mean and the GEFS ensemble mean for both ACC and RMSE at all three lead times. The differences in ACC were significant for all three lead times, and the differences in RMSE were significant at days 3 and 4.

5. Discussion and conclusions

This study used a generative diffusion model to make 1000-member ensembles of integrated water vapor transport (IVT) forecasts over the North Pacific. Once trained, the diffusion model generated a large ensemble at a relatively high speed and low computational cost. The ensembles consisted of high-quality realistic IVT fields that were capable of simulating the structure of observed atmospheric rivers (ARs) that differed from those in the forecast (Fig. 2). Training the model to predict noise rather than directly predict IVT fields can be advantageous because it avoids the double-penalty problem (Chapman et al. 2019). ARs are characterized by sharp gradients of

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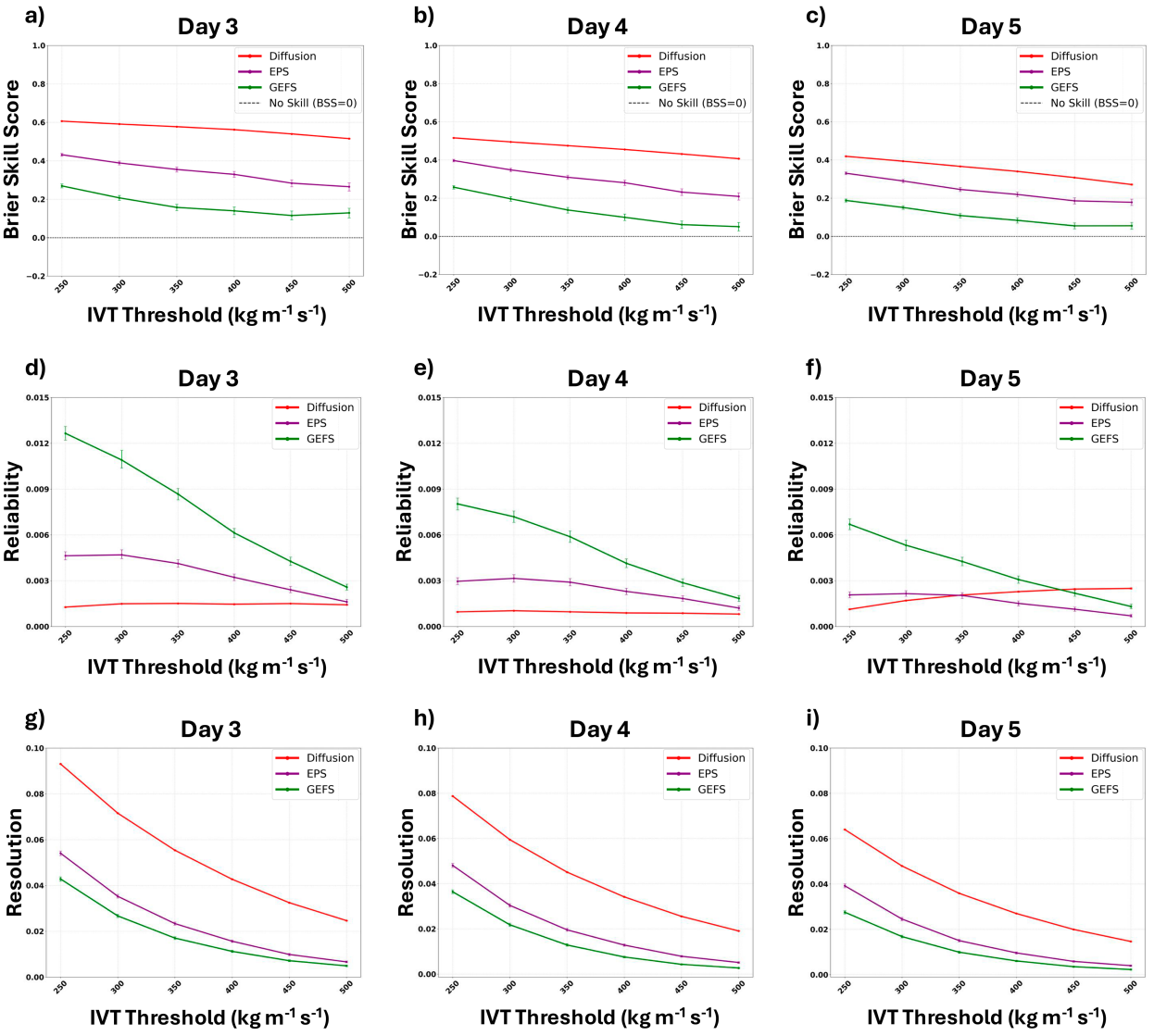

FIG. 8. (a)–(c) Brier skill scores (higher is better) of predictions of IVT exceeding thresholds from 250 to 500 kg m^{-1} s^{-1} in GEFS, EPS, and the diffusion ensemble at lead times of 3–5 days. (d)–(f) Reliability (lower is better) and (g)–(i) resolution (higher is better) are also shown for all three lead times and the same thresholds. Error bars were created from bootstrapping with 1000 bootstraps.

moisture transport, and their prediction accuracy is highly sensitive to the storm location. This can often result in machine learning forecast generation that minimizes a loss function by directly comparing forecasts to observations making predictions of unrealistic blurry fields between possible locations of the AR.

When West-WRF, ERA5, and diffusion ensemble IVT were compared over a single point during an anomalously active AR season, the diffusion ensemble tended to generally capture the uncertainty of the IVT prediction, with ERA5 falling within the 2.5th–97.5th percentile range in 94% of cases on average across lead times (Fig. 3). The misses were slightly asymmetric, with ERA5 exceeding the 97.5th percentile diffusion ensemble member more frequently than it succeeded the 2.5th percentile ensemble member, indicating that the ensemble was slightly underdispersive at the upper tail. This is consistent with the spread–skill relationship (Fig. 6), which shows some underdispersion at relatively high IVT, particularly at the 4- and 5-day lead times. One possible contributing factor for this is that when the input condition (the forecast) does not have an AR or strong IVT, IVT may be restricted to possible states without any AR activity, while input conditions that do have an AR could produce states in which the AR exists in a slightly different location and IVT is low at the fixed point. Despite this underdispersion at the longest lead times, conditions associated with extreme ARs (IVT $>$ 500 kg m^{-1} s^{-1}) were predicted more reliably across the entire spatial domain and test set with the diffusion ensemble than with EPS and GEFS (Figs. 7 and 8), indicating

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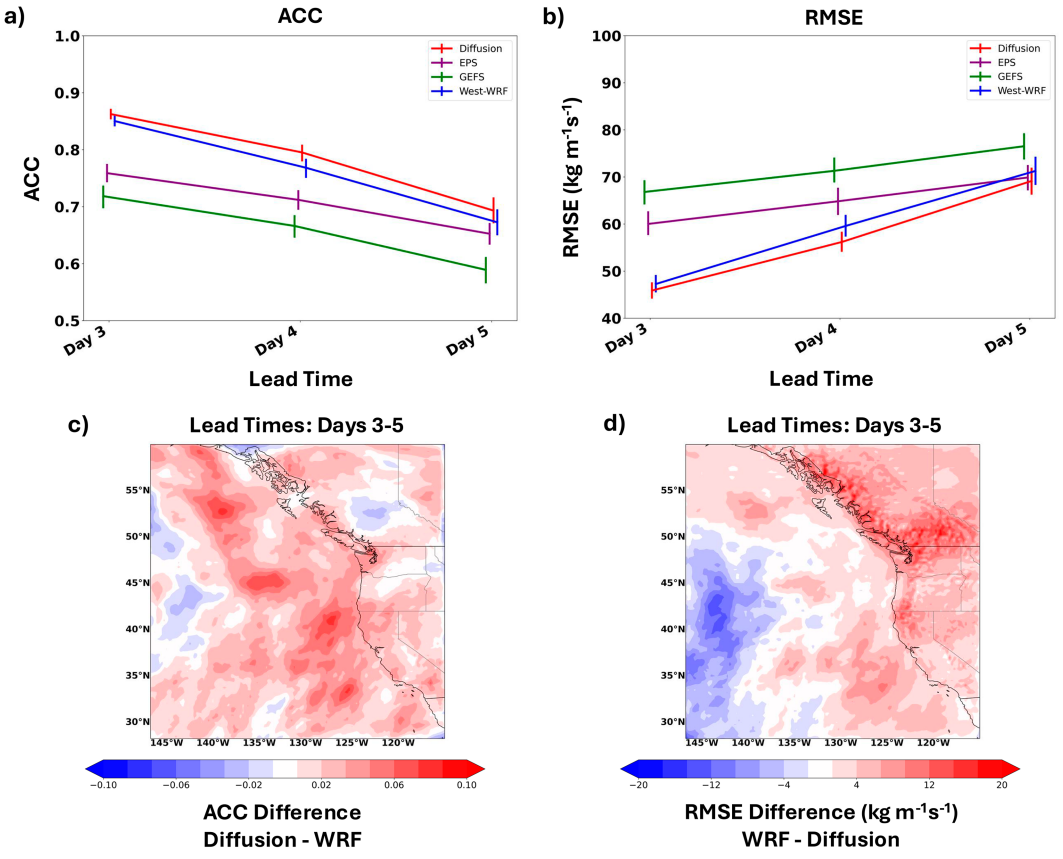

FIG. 9. Deterministic forecast skill of IVT for EPS, GEFS, West-WRF, and the diffusion ensemble mean, measured by (a) ACC and (b) RMSE over the entire domain from days 3 to 5, is shown. Error bars were created from bootstrapping, with 1000 bootstraps and 348 samples. Differences in (c) ACC and (d) RMSE are also shown across the spatial domain for days 3–5. Red indicates locations in which the diffusion ensemble mean had higher skill, and blue indicates locations in which West-WRF had higher skill.

that the diffusion ensemble is a useful tool for forecasting extreme AR conditions relative to the operational baselines.

The diffusion ensemble was also well calibrated overall and improved forecast skill with respect to the EPS and GEFS (Figs. 4–9). While some models may use a probabilistic skill metric as a loss function (e.g., Lang et al. 2024), other models such as this one take approaches that do not. It is critical to note for future experiments that do not use probabilistic loss functions that the calibration skill should be used as a validation metric when selecting the highest-performing model epoch during training if the goal is to produce a reliable ensemble. The ensembles were slightly overdispersive for day 3 and slightly underdispersive for day 4 and day 5. We first tried training the diffusion model with a cosine and sigmoid noise scheduler but found that the linear scheduler was the most effective in reflecting the uncertainty of the system. Linear noise schedulers more aggressively add noise during the early stages of the training process, which could decrease the probability of the samples converging to a precise mode in this context. The calibration and skill could be further improved by continued optimization of noise parameters. Additionally, this framework could be implemented for other variables, lead times, and regions. In particular, diffusion could be applied toward aiding extreme precipitation predictions, which can be captured by AI-based models in the same region (Baño-Medina et al. 2025). Future work could also involve creating autoregressive diffusion ensembles for IVT predictions. While conditioning the diffusion model on a deterministic forecast is highly effective, it is still limited by the skill of the deterministic forecast. If the deterministic forecast makes a poor prediction and the observed state is far from climatology, the diffusion model is prone to struggling. Further improvements in data assimilation techniques, observations, and model physics are therefore also essential for continued progress toward more accurate predictions in the future.

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Data availability statement. The code that can be used to train the diffusion model, run the diffusion model, and create

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the figures can be accessed from the following link: https://github.com/timbhiggins/Diffusion_Ensemble. The pretrained models, diffusion ensemble data, ERA5 data, and West-WRF data can be accessed from the following link: https://scholar.colorado.edu/concern/datasets/3197xn88s. ECMWF and GEFS data were collected using The International Grand Global Ensemble (TIGGE) archives and are hosted by the ECMWF. It can be accessed with the following link: https://ecds.ecmwf.int/datasets/tigge-forecasts?tab=overview.

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